Music Recommendation System

Personalized Discovery at Scale

Million Song Dataset · MIT Applied AI & Data Science Capstone

2M+	**3,337**	**620**	**5**
Data Points	Unique Users	Songs Modeled	Models Built

Problem Definition & Objective

The Business Challenge

In a digital-first world, users face overwhelming music choice. Platforms like Spotify depend on intelligent recommendations to drive engagement and retention.

Without personalization, users churn since they simply cannot discover the music they would love.

Project Objective

- Predict and rank the Top 10 tracks a user is most likely to enjoy
- Automate discovery to maximize platform engagement time
- Evaluate 5 recommendation approaches on the Million Song Dataset
- Identify a production-ready model via RMSE and Precision@K

Framing: Matrix Completion & Ranking: predict missing values in a sparse user × song play-count matrix

Data Overview & Key Insights

2M
interaction records

1M
songs in catalog

138K
rows after filtering

~48%
of songs missing year

Dataset Schema

Field	Source	Description
user_id	count_data	Encoded listener identifier
song_id	both	Encoded track identifier
play_count	count_data	Times a user played a song
artist_name	song_data	Performing artist
title / release	song_data	Track & album name

EDA Insights

- 90% of interactions have play_count ≤ 5 creating a compressed signal range
- Top user played 631 songs; top song played 3,126 times
- Filters applied: ≥90 songs/user, ≥120 listeners/song
- Year excluded as a model feature
- Post-filter: 3,337 users × 620 songs interaction matrix

Solution Design

Five complementary modeling approaches evaluated end-to-end

Popularity Baseline

Avg play_count with minimum listener threshold. Non-personalized cold-start fallback.

User-User Collab Filter

KNNBasic on user similarity vectors. Finds listeners with matching taste profiles.

Item-Item Collab Filter

KNNBasic on song similarity. More stable similarity vectors, but lower Precision@K than user-user in this dataset.

SVD Matrix Factorization

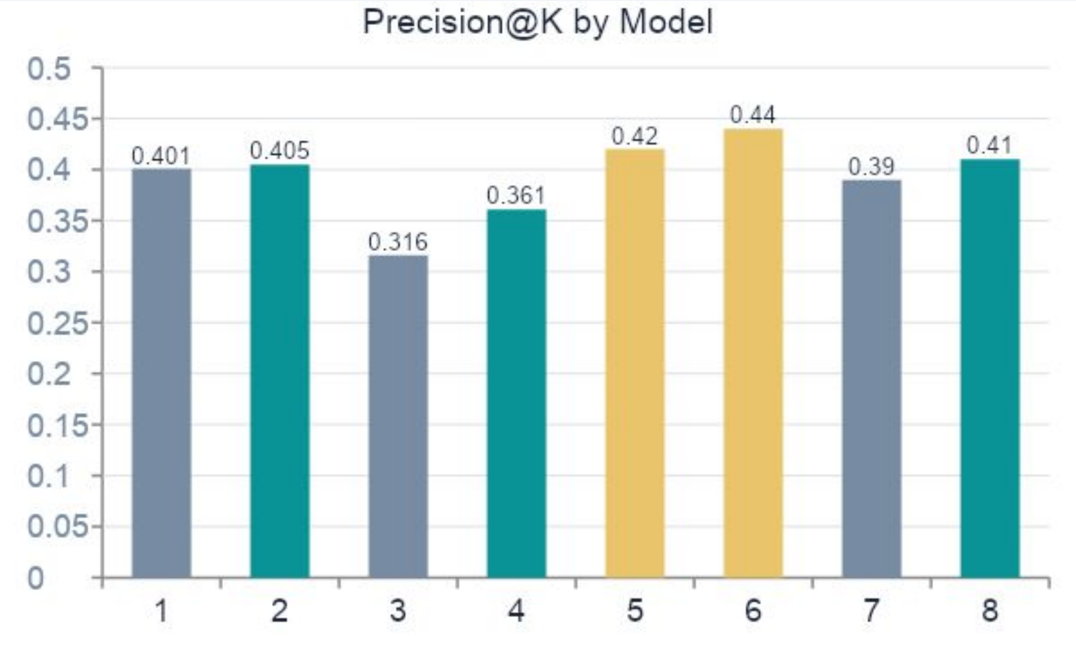
Decomposes interaction matrix into latent factors. Best performer overall.

Content-Based TF-IDF

Cosine similarity on TF-IDF of title + release + artist. Metadata-driven.

+ Hyperparameter tuning (GridSearchCV 3-fold cross-validation) applied to all collaborative models

Model Performance Comparison



RMSE Comparison

Model	Baseline	Tuned
User-User CF	1.082	~1.07
Item-Item CF	1.032	~1.01
SVD ★	**~1.01**	**< 1.00**
Co-Clustering	~1.05	~1.03

Key Takeaway

SVD delivers the best RMSE and Precision@K across all variants. Its latent-factor approach handles sparse matrices better than explicit neighborhood methods.

Data Sparsity Challenge: Even after filtering, most users have interacted with < 1% of available songs. SVD's latent-factor decomposition generalizes across sparse regions giving it a key advantage over KNN methods.

KNN Variant Analysis

Baseline Reference

KNNBasic

Baseline delta:

Precision: 0.401 | Recall: 0.705

Tuned delta:

Precision: 0.405 | Recall: ~0.71

Raw cosine similarity on play counts. No bias correction. Surprisingly competitive given the compressed play_count range.

Solid baseline

Mean Centering

KNNWithMeans

Baseline delta:

Precision: −0.009 | Recall: −0.121

Tuned delta:

(vs. KNNBasic baseline)

Subtracts each user's average before similarity. Hurt performance. User means are too similar in a 1-5 range for centering to add signal.

Underperformed

Full Bias Correction

KNNBaseline

Baseline delta:

Precision: +0.010 | Recall: −0.048

Tuned delta:

Precision: −0.006 | Recall: −0.007

Corrects for global avg, user bias, and item bias simultaneously. Best untuned KNN variant but converges with KNNBasic after tuning.

Best KNN variant

Key Conclusion

All three KNN variants converge to the same performance ceiling after tuning. When Basic, WithMeans, and Baseline all land at the same score regardless of how sophisticated the similarity calculation is, the bottleneck is the data, not the algorithm. This is the strongest evidence that SVD's fundamentally different approach is the right choice.

SVD Matrix Factorization — Deep Dive

How It Works

1. Decomposes the user × song matrix into two smaller matrices; a user latent factor matrix and a song latent factor matrix
2. Each user and song is represented as a vector of hidden (latent) characteristics learned from play_count patterns
3. A predicted play_count is estimated by taking the dot product of a user vector and a song vector
4. Bias terms capture global average, user tendency, and song popularity independently

Key Hyperparameters (Tuned)

Parameter	Best Value	Controls
n_factors	100	Dimensions of latent space
n_epochs	20	Training iterations over the data
lr_all	0.005	How fast the model learns (step size)
reg_all	0.02	Penalty to prevent overfitting

< 1.00
RMSE (Tuned)
Best across all models

0.440
Precision@K
vs 0.405 User-User tuned

0.313
Log-Norm RMSE
Best single improvement

Why SVD Outperforms KNN on This Dataset

Tuning headroom: User-User precision plateaus (0.401→0.405); SVD improves meaningfully (0.420→0.440+)

Generalization: Latent factors learned from the full matrix. Stable across sparse user histories where KNN neighbor pools thin out

Content-Based Filtering & Hybrid Potential

How Content-Based Works

1. Concatenate title + release + artist_name into a text feature
2. Tokenize, remove stopwords, and lemmatize with NLTK
3. Vectorize with TF-IDF (sklearn)
4. Compute cosine similarity between all song pairs
5. Return top-10 most similar songs for a given title

Example: "Learn To Fly" (Foo Fighters)

Returned other Foo Fighters tracks and artists with descriptively similar metadata confirming artist-name dominance in the TF-IDF vector.

Strengths & Limitations

- No play-count required and works for new songs
- "More like this" discovery use case
- Interpretable: artist / album / title signals
- Artist name dominates. The model becomes artist recommender
- Blind to actual listener engagement
- Cannot personalize to individual users

Hybrid Opportunity

Combining SVD collaborative scores with content-based similarity as a re-ranking layer would provide both personalization depth and the ability to surface cold-start songs. This is the recommended next-step architecture.

Refined Insights

📉 Compressed Signal

~90% of interactions have play_count ≤ 5. This tight range limits differentiation. Every model's RMSE clusters near 1.0 as a result.

🌐 Sparsity is the Core Challenge

After aggressive filtering, most users have still interacted with < 1% of songs. This is why neighborhood methods struggle and SVD excels.

📅 Year Data Unreliable

~48% of songs carry year = 0 (missing). Correctly excluded from all models; retained only as optional metadata for post-retrieval filtering.

🎤 Artist Dominates Content Similarity

TF-IDF feature engineering made the content model an artist-recommender. Future work could separate artist, title, and genre into weighted fields.

📊 Popularity Ranking Requires Care

Raw average play_count distorts rankings. A minimum-listener threshold, or Bayesian average, is essential for meaningful popularity signals.

🔗 Cross-Model Agreement = Confidence

Songs appearing in top-5 lists across SVD, KNN, and co-clustering represent high-confidence recommendations. This is a strong ensemble signal.

Recommendations & Next Steps

★ Recommended Production Model: Tuned SVD (Matrix Factorization)

Best RMSE and Precision@K across all models. Resilient to the sparse interaction matrix because latent factors generalize across unobserved user-song pairs giving it a key advantage where explicit neighbors are unreliable.

Recommended Model Roles

SVD (primary)	Personalized homepage & queue recommendations
Popularity model	Cold-start fallback for new / inactive users
Content-based	"More like this" / similar-song feature
Co-Clustering	Ensemble diversity complement to SVD

Next Steps

- Log normalization confirmed RMSE improvement (1.01 → 0.32). Bayesian normalization and play_count cap relaxation remain as further avenues
- Build hybrid model: SVD scores + content similarity re-ranking
- Explore SVD++ — extends SVD by incorporating implicit feedback signals, more theoretically aligned with play-count data
- Enrich content-based features: separate artist, title, and genre into weighted TF-IDF fields to reduce artist-name dominance

Thank You

Music Recommendation System
Million Song Dataset Capstone

Dataset: 2M interactions → 138K filtered records

Models: Popularity · User-CF · Item-CF · SVD · Content-Based

Best Model: Tuned SVD with the best RMSE & Precision@K

Next Step: Hybrid SVD + Content re-ranking layer

Questions?